

Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

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Perturbation evidence is multiplying faster than it can be organised

A single genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, while pooled screens test thousands of targets. That scale could reveal robust biology across studies—but identifiers, metadata, assays and processing pipelines differ. Results remain hard to compare or reproduce. The goal of the Perturbation Catalogue is to supply a shared schema, stable gene identity and one access path for combining this evidence.

36× growth
in perturbation papers
over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: Replogle et al., Cell 2022.

Our approach turns scattered datasets into a unified resource

1,237

datasets

19,235

targets

36

tissues

34

cell types

1,200

cell lines

169

diseases

CRISPR screens · 1,201 datasets

Pooled guide libraries perturb genes at scale. Phenotypes and fitness measurements reveal essentiality and context-specific dependencies.

Perturb-Seq · 26 datasets

Guide identity is paired with single-cell RNA-seq, linking perturbations to cell states and gene-regulatory responses.

MAVE · 10 datasets

Variant libraries and functional assays measure sequence-to-function effects across many alleles, making subtle molecular changes searchable.

We extensively curate metadata...

84

meta-data fields

Study & provenance

Title, authors, year, URI, licence, linked datasets

Perturbation & library

Targets, library design, delivery, guides, variants

Model system

Organism, tissue, cell type or line, disease, sex, stage

Readout & processing

Assay, technology, sequencing kit, reference

Stable Ensembl gene IDs and ontology-linked entities make the same biological question addressable across modalities.

And reprocess data from scratch

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected Perturb-Seq studies follow one shared computational path.

SRA / FASTQ

Raw sequencing reads and study-specific probe lists

→

Cell × gene counts

Shared reference and gene annotation

→

Guide assignment

Map cells to perturbations

→

QC + filtering

Produce the final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA.

Browse the Catalogue

Target search, current coverage and biological-context summaries in one human-facing portal.

Access records via API

Search and retrieve metadata or modality results through the same backend as the portal.

GET /v1/perturb-seq/nadig_2025_jurkat/search?perturbation_gene_name=TFAM&limit=1

```
{
  "perturbation": {
    "gene_symbol": "TFAM"
  },
  "effect": {
    "gene_symbol": "MT-ND5",
    "log2fc": -1.926,
    "padj": 0.0
  }
}
```

Download entire datasets

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

Stay tuned for more exciting things coming soon

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Reanalyse MAVE data and enrich variant-to-function evidence

Extend functional-variant coverage with harmonised measurements, scores and enrichment-ready outputs.

Explore: ebi.ac.uk/perturbation-catalogue

Get in touch: perturbation-catalogue-help@ebi.ac.uk